

Math 328 Number Theory

Personal revision notes I made for the LU 2025-2026 Math 321 exams.

Revision Checklist:

- ☐ Solve linear Diophantine equations
- ☐ Simple proofs: Characterisation of solutions
- ☐ Named theorems: Wilson's, Fermat's, Chinese Remainder, Law of Quadratic Reciprocity
- ☐ Solve simultaneous congruence equations using the Chinese Remainder Theorem
- ☐ Calculate Legendre Symbol
- ☐ Solve quadratic congruence equations
- ☐ Calculate reduced form of a BQR
- ☐ Calculate continued fraction expansion
- ☐ Solve Pell equations, remember characterisation of all solutions

Chapter 1: Arithmetic

- **Division Algorithm** Let $n/m \in \mathbb{Z}$ with $m > 0$. Then, there exist unique $q, r \in \mathbb{Z}$ such that $n = qm + r$ and $0 \leq r < m$.
- Use Euclidean's Algorithm to compute gcd of two integers.
- **Diophantine Equation** A polynomial equation with integer coefficients that we wish to find integer solutions of.
- How to find solutions to Diophantine Equation $ax + by = c$
 1. Run Euclidean's algorithm on a and b to calculate their gcd
 2. Check gcd divides c
 3. Reverse Euclidean algorithm
 4. multiply both sides by a factor to get c on one side

- How to find all solutions to a Diophantine Equation $ax + by = c$:
 1. Find an integer solution (x_0, y_0) using previous steps. All solutions are given by $(x_0 + \frac{b}{\gcd(a,b)}k, y_0 - \frac{a}{\gcd(a,b)}k)$ for all $k \in \mathbb{Z}$
- **Fundamental Theorem of Arithmetic** Every positive integer has a prime factorisation, unique up to the order of primes

Chapter 2: Modular Arithmetic

- Let $N \geq 1$. For any $\bar{a}, \bar{b} \in \mathbb{Z}/N\mathbb{Z}$
 - $\overline{a+b} = \bar{a} + \bar{b}$
 - $\overline{ab} = \bar{a}\bar{b}$
 - $\overline{a^n} = \bar{a}^n$
- How to find solutions $x \in \mathbb{Z}/N\mathbb{Z}$ to $ax \equiv 1 \pmod{N}$, ie., a^{-1} the multiplicative inverse of a modulo N
 1. Use theorem: The congruence $ax \equiv b \pmod{N}$ has a solution iff $\gcd(a, N) | b$. So a has an inverse iff a and N are coprime.
- How to find solutions to a general congruence equation $ax \equiv b \pmod{N}$
 1. Find a^{-1}
 2. Multiply both sides by a^{-1}
- **Multiplicative Group Modulo N** Group consisting of a subset of $\mathbb{Z}/N\mathbb{Z}$ that has a multiplicative inverse modulo N .
- **Wilson's Theorem** For p prime, $(p-1)! \equiv -1 \pmod{p}$
- **Fermat's Little Theorem** Let p be prime and $a \in \mathbb{Z}$. If a and p are coprime then $a^{p-1} \equiv 1 \pmod{p}$. (The powers of $a \pmod{p}$ repeat every $p-1$)
Use Wilson's theorem to prove this.
- **Chinese Remainder Theorem** Suppose $N_1, N_2, \dots, N_m \geq 1$ are pairwise coprime and that $a_1, a_2, \dots, a_m \in \mathbb{Z}$. Then there exists unique a such that

$$x \equiv a \pmod{N_1 \cdots N_m} \iff x \equiv a_i \pmod{N_i}$$

for all i

- How to find such a ?
 1. Write $S_i = \prod_{j \neq i} N_j$ and $b_i = S_i^{-1} \pmod{N_i}$
 2. Let $a = \sum_{i=1}^m S_i b_i a_i$

Chapter 3: Quadratic Residues

- **Quadratic Residue** Let $N \geq 1$. Then $a \in \mathbb{Z}/N\mathbb{Z}$ is a quadratic residue if $(\exists b \in \mathbb{Z}/N\mathbb{Z})(b^2 = a)$

We study QR mod primes

R_p denote the set of non-zero quadratic residues in $\mathbb{Z}/p\mathbb{Z}$ and S_p denote the set of quadratic non-residues

- n is a QR mod N iff n is a QR mod p for all primes p dividing N .
- **Legendre Symbol** Let p be an odd prime and $a \in \mathbb{Z}$ be such that a and p are coprime. Then, the Legendre symbol $\left(\frac{a}{p}\right) \in \{\pm 1\}$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & a \in R_p, \\ -1 & a \in S_p. \end{cases}$$

- Properties of the Legendre Symbol

$$\begin{aligned} - a \equiv b \pmod{p} &\implies \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right) \\ - a, b \in \mathbb{Z} &\implies \left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) \end{aligned}$$

- **Euler's Criterion** Let p be an odd prime and $a \in \mathbb{Z}$ such that a and p are coprime. Then,

$$a^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right) \pmod{p}$$

- **Quadratic Reciprocity** Let $p, q \geq 1$ be odd primes. Then

$$\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}} \left(\frac{q}{p}\right)$$

- How to calculate $\left(\frac{a}{p}\right)$

1. factorise a into a product of primes $\pm \prod_i p_i^{e_i}$.
2. Calculate $\left(\frac{-1}{p}\right)$.
3. Calculate $\left(\frac{p_i}{p}\right)$ for all i
4. Use property 2 of legendre symbols
5. We can use:

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & p \equiv 1, 7 \pmod{8} \\ -1 & p \equiv 3, 5 \pmod{8} \end{cases}$$

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & p \equiv 1 \pmod{4} \\ -1 & p \equiv 3 \pmod{4} \end{cases}$$

– Quadratic Reciprocity

- **Solvability of quadratic congruences**

Let $a, b, c \in \mathbb{Z}$. Suppose $\gcd(2a, N) = 1$. The congruence $ax^2 + bx + c \equiv 0 \pmod{N}$ has a solution iff for each prime power $p_i^{e_i} | N$ and $p_i^{e_i+1} \nmid N$ and $\Delta = b^2 - 4ac$ satisfying $\Delta = p_i^{a_i} \Delta_i$ for some a_i (ie., $p_i | \Delta$) satisfy either:

- $a_i \geq e_i$ or
- a_i is even and $\left(\frac{\Delta_i}{p_i}\right) = 1$

Note that a_i can be 0 in which case, we check whether $\left(\frac{\Delta}{p_i}\right) = 1$.

Chapter 4: Some quadratic Diophantine Equations

- **Binary Quadratic Form** Polynomial of the form

$$f(x, y) = ax^2 + bxy + cy^2$$

- **Primitive Solution** Any solution to $f(x, y) = n$ with $\gcd(x, y) = 1$ is called a primitive solution.

- **Equivalent BQF's** Two BQF's are equivalent if one is a linear transformation of the other and the determinant of the change of matrix transformation is ± 1
- **Properly equivalent** if the change of matrix transformation is 1
- Equivalent BQF's (primitively) represent the same integers
- **Discriminant** $\Delta = b^2 - 4ac$ when $f(x, y) = ax^2 + bxy + cy^2$
- BQF is positive definite iff $\Delta < 0$ and $a, c > 0$.
- Let Δ be negative with $\Delta \equiv 0, 1 \pmod{4}$. Then $n \in \mathbb{Z}_{>0}$ is primitively represented by some positive definite BQF of discriminant Δ iff $\Delta \in \mathbb{Z}/4n\mathbb{Z}$ is a quadratic residue
- **Reduced Form of a BQF** Let $f(x, y) = ax^2 + bxy + cy^2$ be a positive definite BQF. $f(x, y)$ is in reduced form iff:
 1. $|b| \leq a \leq c$
 2. $b \geq 0$ whenever $a = |b|$ or $a = c$
- Gauss: Every positive definite binary quadratic form is properly equivalent to a unique reduced form
- Algorithm to find the reduced form of a BQF:
 1. Define the transformations: (these transformations are proper since the determinant of the change of variables matrix is 1.)

$$S : (x, y) \mapsto (-y, x)$$

$$T_n : (x, y) \mapsto (x - ny, y)$$

2.
 - Case 1: $|b| > a$. Perform T_n . Choose n such that the transformed values of the coefficients satisfy $|b| > a$
 - Case 2: $a > c$. Perform S .
 - Case 3: $|b| \leq a \leq c$: with $b < 0$. If $a = |b|$ then perform T_{-1} . If $a = c$ perform S
- **Class Set** $Cl(\Delta)$ set of positive definite reduced forms of discriminant Δ
 - Any negative Δ has finite class set.

Chapter 5: Continued Fractions

- **(Simple) Continued Fraction** Possibly infinite fraction of the form

$$[a_0; a_1; a_2; \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\dots}}}$$

- **nth convergent** The nth convergent of $\alpha = [a_0; a_1; a_2; \dots]$ is

$$C_n = \frac{p_n}{q_n} = [a_0; a_1; \dots a_n]$$

- **Pell's Equation** A Diophantine of the form $x^2 - dy^2 = 1$ for a fixed non-square $d > 0$
- Any solution (p, q) of a Pell's equation $x^2 - dy^2 = 1$ has $\frac{p}{q}$ a convergent of continued fraction of $\sqrt{d}$ Converse is not true.
- Important Theorem to find all solutions to a Pell's equation

A continued fraction is eventually periodic iff it represents a number of the form $\alpha = a + b\sqrt{d}$ for some $a, b \in \mathbb{Q}$ and some non-square integer $d > 0$.

Let $d > 0$ be a non-square integer and n be the period of the continued fraction of $\sqrt{d}$. Then, the positive solutions to $x^2 - dy^2 = 1$ are given by the following convergents for $k \geq 1$

$$(x, y) = \begin{cases} (p_{kn-1}, q_{kn-1}) & n \equiv 0 \pmod{2} \\ (p_{2kn-1}, q_{2kn-1}) & n \equiv 1 \pmod{2} \end{cases}$$

- Easier way to find all solutions if you know one positive solution

Let $d > 0$ be a non-square integer and suppose that (a_0, b_0) is the smallest positive solution to $x^2 - dy^2 = 1$. Then all positive solutions (x, y) are given by the coefficients of $(a_0 + b_0\sqrt{d})^m$ for all positive integer m . (There are infinitely many solutions)

- How to find solutions to a Pell's equation $x^2 - dy^2 = 1$:
 1. Compute continued fraction expansion of $\sqrt{d}$
 2. Either use first theorem or
 3. Compute convergents until we get a solution to the equation then use second theorem to find the rest of the solutions.